

Appendix: Artifact Description/Artifact Evaluation

Artifact Description (AD)

Paper ID: pap142

Requested SC26 badge review: all three artifact badges.

I. CONTRIBUTIONS AND ARTIFACTS

A. Main Contributions

- C1: TDPs expose target-centric temporal dependencies and branch-level parallelism.
- C2: DDTC constructs TDPs on demand to reduce irrelevant work and data movement.
- C3: OATS and PHLE reuse overlapping dependency packets while preserving ordered updates.
- C4: U280 evaluation across six datasets and four TGNN models, including accelerator-baseline comparisons.

B. Computational Artifacts

TABLE I
ARTIFACT-TO-RESULT MAP.

ID	Contents	Contributions and results
A1	Four U280 sources, xclbins, hosts, inputs, tests, and measurement workflow	C1–C4; U280 latency behavior
A2	results/result.csv and the CSV/SVG generator	C1–C4; Fig.2, Fig.4(a), Fig.9(b), and Fig.10-Fig.14

Artifact: [10.5281/zenodo.21417187](https://zenodo.org/record/21417187); source: [TempGNN-Program/TempGNN](#); tag: sc26-ae-pap142-v1.0.

II. DOWNLOAD

```
git clone https://github.com/TempGNN-Program/
↪ TempGNN.git
cd TempGNN
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

III. ENVIRONMENT AND INPUTS

Board: AMD/Xilinx Alveo U280,
↪ xcu280-fsvh2892-2L-e
Platform: xilinx_u280_gen3x16_xdma_1_202211_1
OS: Ubuntu 22.04 x86_64
XRT: 2.16.204
Vitis/Vivado: 2023.2 (xclbin rebuild only)
GCC: 11.4.0
GNU Make: 4.3
Python: 3.10 or newer

XRT: <https://github.com/Xilinx/XRT>. Python: <https://www.python.org/>. A reviewer account on the matching U280 host is provided through the private SC26 submission channel.

A1 uses bundled WK, MC, RT, LM, WT, and GT prefixes with JODIE, TGN, TGAT, and APAN. Metadata and hashes are under external/u280_dataset_samples/. A2 uses results/result.csv.

IV. RUN ON U280

A1 supports C1–C4.
source /opt/xilinx/xrt/setup.sh
make ae-core-u280 U280_CORE_DEVICE=0
↪ U280_CORE_REPETITIONS=3
1) check four U280 artifacts;
2) prepare six-dataset/four-model inputs;
3) run four implementations three times;
4) write raw CSVs; and
5) aggregate mean latency.

TABLE II
EXPECTED U280 MEAN LATENCY.

Implementation	Latency (ms)
TempGNN	1.296553
MATG	9.966618
ViTeGNN	2.869752
RTGA	4.192824

A successful run exits with status zero, records all four implementations, and writes results/reviewer_u280_runs/<run-id>/. TempGNN should retain the lowest mean latency near 1.3 ms. Expected time: 10–20 min setup, 30–90 min execution, and less than 5 min analysis.

V. GENERATE FIGURES FROM RESULT.CSV

A2 supports C1–C4.
python3 -m scripts.reproduce_paper_figures
The command reads results/result.csv and generates nine CSV/SVG pairs for Fig.2, Fig.4(a), Fig.9(b), and Fig.10-Fig.14 under results/paper_reproduction/. The manifest must identify results/result.csv as the input. Expected time: less than 5 min setup, less than 1 min execution, and 2–5 min analysis.

VI. BASELINE STATUS

- MATG: independently reproduced from the IPDPS 2022 paper and its partial public HLS headers; includes a distinct source, runner, and U280 xclbin.
 - ViTeGNN: independently reproduced from the TPDS 2025 paper; includes a distinct source, runner, and U280 xclbin.
 - RTGA: independently reproduced from the DAC 2024 paper; includes a distinct source, runner, and U280 xclbin.
 - Cascade and TGLite-CPU: retained only as packaged paper comparison records; no fresh execution of those external stacks is claimed.
 - TempGNN: includes HLS kernels, testbenches, Vitis build scripts, XRT host code, U280 board logs, and its own U280 xclbin.
- The three baselines are independent, paper-based forward-path reproductions, not the authors' complete original stacks.

VII. METRIC MEANING

DDTC is dependency-driven TDP construction. OATS is overlap-aware TDP synchronization.
packet_reuse_factor measures how many per-target packet materializations collapse into unique PHLE packets. memory_bytes is packet state/metadata traffic. cycles is the kernel's cycle proxy used by C-sim and exported fixture stats. HLS reports provide C/RTL latency after synthesis/cosim.

VIII. LICENSE

This artifact is licensed under the Apache License 2.0. See LICENSE.

Artifact Evaluation (AE)

Paper ID: pap142

This workflow evaluates A1 (four U280 implementations) and A2 (results/result.csv and the figure generator).

Artifact: [10.5281/zenodo.21417187](https://zenodo.org/record/21417187); source: [TempGNN-Program/TempGNN](#); tag: sc26-ae-pap142-v1.0.

I. DOWNLOAD AND SETUP

```
git clone https://github.com/TempGNN-Program/
└─ TempGNN.git
cd TempGNN
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
Board: AMD/Xilinx Alveo U280,
└─ xcu280-fsvh2892-2L-e
Platform: xilinx_u280_gen3x16_xdma_1_202211_1
OS: Ubuntu 22.04 x86_64
XRT: 2.16.204
Vitis/Vivado: 2023.2 (xclbin rebuild only)
GCC: 11.4.0
GNU Make: 4.3
Python: 3.10 or newer
```

The U280 account and login instructions are delivered through the private SC26 submission channel.

II. REVIEW PLAN

TABLE III
EXPECTED DIRECT REVIEW TIME.

ID	Setup	Run	Analysis	Expected result
A1	10–20	30–90	<5	Four fresh results; TempGNN lowest near 1.3 ms
A2	<5	<1	2–5	Nine pairs from results/result.csv

All times are minutes. Optional xclbin rebuilding is excluded.

III. EXECUTE A1

```
source /opt/xilinx/xrt/setup.sh
xrt-smi examine
make ae-core-u280 U280_CORE_DEVICE=0
└─ U280_CORE_REPETITIONS=3
```

- 1) check four packaged U280 artifacts;
- 2) load six datasets and four models;
- 3) run each implementation three times;
- 4) write raw measurement CSVs; and
- 5) aggregate latency and update the AE report.

Datasets: WK, MC, RT, LM, WT, GT
Models: JODIE, TGN, TGAT, APAN
Repetitions: 3

Fresh results are under results/reviewer_u280_runs/<run-id>/.

IV. ANALYZE A1

Success requires a zero exit status; fresh rows for all four implementations; all six datasets and four models; and the expected behavior in Table IV. The principal behavior is that TempGNN has the lowest mean latency near 1.3 ms, supporting C1–C4.

TABLE IV
EXPECTED U280 MEAN LATENCY.

Implementation	Latency (ms)
TempGNN	1.296553
MATG	9.966618
ViTeGNN	2.869752
RTGA	4.192824

V. EXECUTE AND ANALYZE A2

```
python3 -m scripts.reproduce_paper_figures
```

- 1) read results/result.csv;
- 2) validate and group rows by paper figure;
- 3) generate a CSV and SVG for each figure; and
- 4) write figure_data_manifest.csv.

Expected output is nine CSV/SVG pairs covering Fig.2, Fig.4(a), Fig.9(b), and Fig.10-Fig.14. Success requires all pairs and a manifest that records results/result.csv as their input.

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- Cascade and TGLite-CPU: retained only as packaged paper comparison records; no fresh execution of those external stacks is claimed.
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